

Programmazione Concorrente, Parallela e su Cloud
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Amazon Web Services:
Management

Carmine Spagnuolo
2018

Outline

AWS Management Introduction

Amazon EC2 Command Line Interface

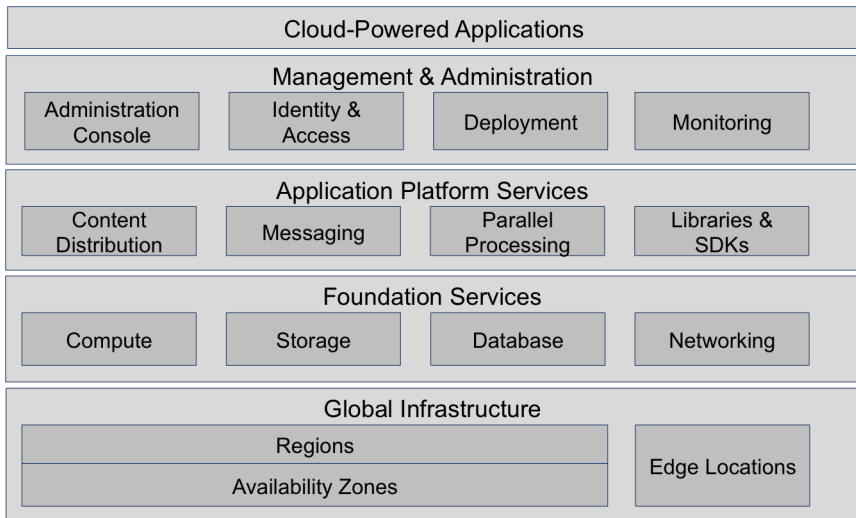
Amazon EC2 Web Console

StarCluster cluster-computing toolkit for Amazon EC2

AWS Management Introduction

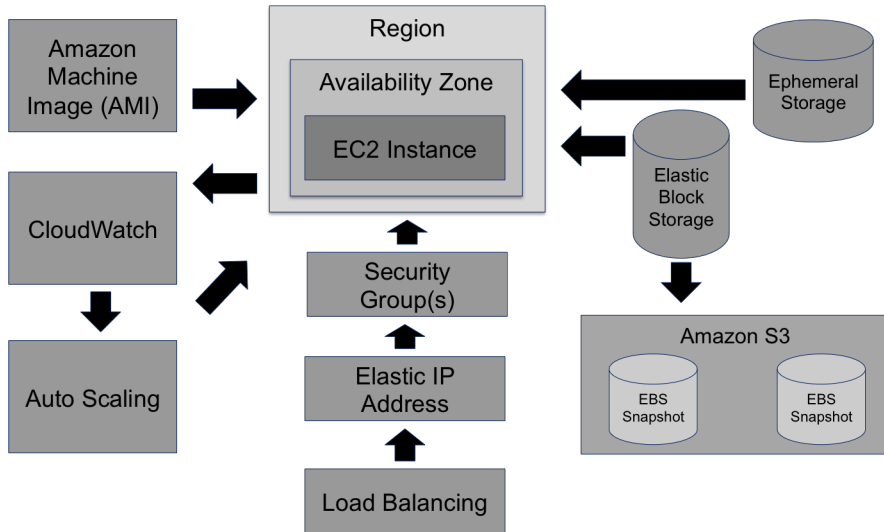
AWS Management

Architecture



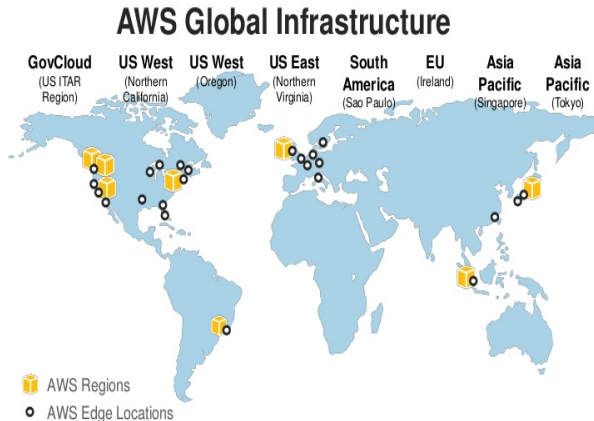
AWS Management

Architecture



AWS Management

Regions and Availability Zones



► 7 Regions

► Availability Zones:

- Physical infrastructure – 1 or more data centers;
- 2 or more AZ's per Region;
- Fault tolerance across AZ's.

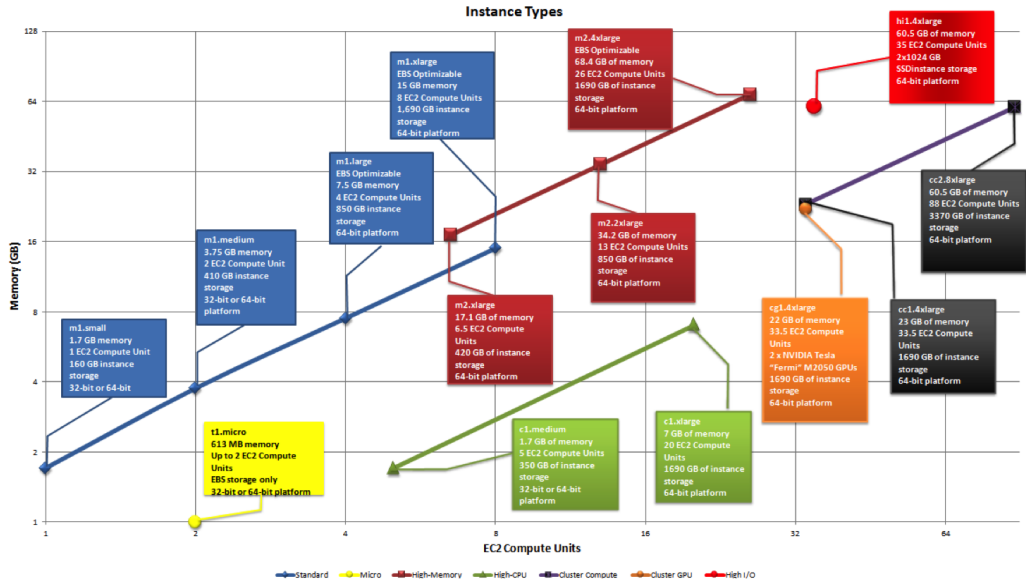
AWS Management

Security Group and Elastic IP Address

- ▶ **Security group** acts as a virtual firewall that controls the traffic for one or more instances:
 - ▶ each instances have one or more security groups;
 - ▶ rules defines the traffic allowed by a security group;
 - ▶ rules can be modified at any time; new rules are automatically applied to all instances that are associated with the security group.
- ▶ **Elastic IP address** is a static IPv4 address designed for dynamic cloud computing:
 - ▶ is associated on an AWS account;
 - ▶ is possible to easily manage a failure of an instance or software by rapidly remapping the address to another instance.

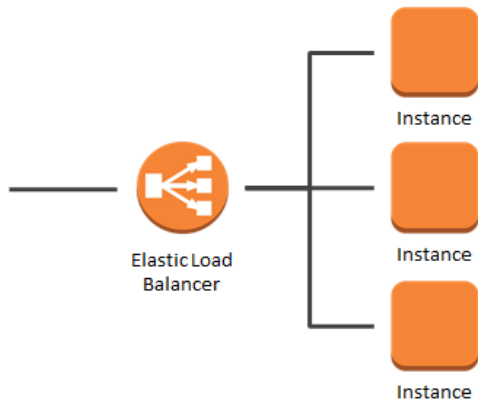
AWS Management

EC2 Instances



AWS Management

EC2 Elastic Load Balancer



- ▶ ELB automatically distributes incoming application traffic across multiple Amazon EC2 instances.
- ▶ Enables the user to achieve fault tolerance in applications.

AWS Management

EC2 Auto Scaling



- ▶ Allows the user to scale on Amazon EC2 capacity up or down automatically according to defined conditions.
- ▶ Automatically increases the number of Amazon EC2 instances according to the demanding.
- ▶ Based on demanding patterns or experience hourly, daily, or weekly variability in usage.

AWS Management

EC2 Auto Scaling

- ▶ Currently no direct console UI is available:
 - ▶ Command Line Interface;
 - ▶ API;
 - ▶ Elastic Beanstalk, that is an easy-to-use service for deploying and scaling web applications and services developed with Java, .NET, PHP, Node.js, Python, Ruby, Go and on familiar servers such as Apache, Nginx.

AWS Management

Access to Amazon Web Services Infrastructure

- ▶ Amazon EC2 Command Line Interface.
- ▶ Amazon EC2 Web Console.

Amazon EC2 Command Line Interface

Amazon EC2 CLI

What is Amazon EC2 CLI

AWS Command Line Interface (CLI):

- ▶ unified tool to manage your AWS services;
- ▶ allows the user to download and configure multiple AWS services from the command line and automate them through scripts.

AWS security credentials:

- ▶ allows to verify who you are and whether you have permission to access the resources that you are requesting;
- ▶ AWS uses the security credentials to authenticate and authorize your requests.

Amazon EC2 CLI

Install Amazon EC2 CLI

- ▶ Reference: <https://goo.gl/YnyAvJ>
- ▶ AWS Security Credentials, Account Identifiers: <https://goo.gl/TUYMQU>
- ▶ Simple way to install using Python:
 - ▶ **`curl "https://bootstrap.pypa.io/get-pip.py" -o "get-pip.py"`**
 - ▶ **`sudo python get-pip.py`**
 - ▶ **`sudo pip install awscli`**

[<http://docs.aws.amazon.com/cli/latest/userguide/tutorial-ec2-ubuntu.html>]

Amazon EC2 CLI

Configure the CLI and Launch an EC2 Instance

Execute in your command line the command "aws configure":

- ▶ AWS Access Key ID [None]: AKIAIOSFODNN7EXAMPLE
- ▶ AWS Secret Access Key [None]: wJalrXUtnFEMI/K7MDENG/bPxRfiCYEXAMPLEKEY
- ▶ Default region name [None]: us-east-1
- ▶ Default output format [None]: json

[Find your Credentials on Access Keys (Access Key ID and Secret Access Key)]

Amazon EC2 CLI

Create a Security Group, Key Pair, and Role for the EC2 Instance

Create a Security Group

```
aws ec2 create-security-group --group-name devenv-sg --description  
"security group for development environment in EC2"
```

Authorize a Security Group for SSH connection

```
aws ec2 authorize-security-group-ingress --group-name devenv-sg  
--protocol tcp --port 22 --cidr 0.0.0.0/0
```

Amazon EC2 CLI

Create a Security Group, Key Pair, and Role for the EC2 Instance

Create and Download SSH credential

```
aws ec2 create-key-pair --key-name devenv-key --query 'KeyMaterial'  
--output text > devenv-key.pem
```

Change permission in the way that only you have access to the key file

```
chmod 400 devenv-key.pem
```

Amazon EC2 CLI

Launch and Connect to the Instance

Run an instance of Ubuntu Server 16.04 LTS – don't forget to change the security group

```
aws ec2 run-instances --image-id ami-f4cc1de2 --security-group-ids  
sg-98790ee7 --count 1 --instance-type t2.micro --key-name devenv-key  
--query 'Instances[0].InstanceId'
```

Get the IP of the running instance

```
aws ec2 describe-instances --instance-ids i-ec3e1e2k --query  
'Reservations[0].Instances[0].PublicIpAddress'
```

Amazon EC2 CLI

Launch and Connect to the Instance

Connect to the instance by SSH

```
ssh -i devenv-key.pem ubuntu@54.236.37.209
```

Amazon EC2 Web Console

Amazon EC2 Web Console

- ▶ [AWS Console Home](#)

AWS services

Find a service by name (for example, EC2, S3, Elastic Beanstalk).

Recently visited services

 [Support](#)

 [IAM](#)

 [Billing](#)

 [EC2](#)

> [All services](#)

Build a solution

Get started with simple wizards and automated workflows.

 [Launch a virtual machine](#)
With EC2
~1 minute

 [Build a web app](#)
With Elastic Beanstalk
~5 minutes

 [Deploy a serverless microservice](#)
With Lambda, API Gateway
~2 minutes

 [Host a static website](#)
With S3, CloudFront, Route 53
~5 minutes

 [Create a backend for your mobile app](#)
With Mobile Hub
~5 minutes

 [Register a domain](#)
With Route 53
~3 minutes

Learn to build

Learn to deploy your solutions through step-by-step guides, labs, and videos.

[See all](#)

Websites



3 videos, 3 tutorials, 3 labs

DevOps



6 videos, 2 tutorials, 3 labs

Backup and recovery



3 videos, 2 tutorials, 3 labs

Big data



3 videos, 2 tutorials, 3 labs

Databases



3 videos, 5 tutorials, 3 labs

Mobile



3 videos, 1 lab

Helpful tips



Manage your costs

Get real-time billing alerts based on your cost and usage budgets. [Start now](#)



Create an organization

Use AWS Organizations for policy-based management of multiple AWS accounts. [Start now](#)

What's new?

Announcing Amazon Chime

Learn how this new communication service makes it easy for employees to communicate with voice, video and chat. [Learn more](#)

Introducing Elastic Volumes for Amazon EBS

Learn how this new capability allows you to modify configurations of live volumes with a simple API call or a few console clicks. [Learn more](#)

[See all](#)

AWS Marketplace

Discover, procure, and deploy popular [software products](#) that run on AWS.

Have feedback?

[Submit feedback](#) to tell us about your experience with the AWS Management Console.

Amazon EC2 Web Console

- ▶ AWS Console Home
- ▶ AWS Services

History

Console Home

Support

IAM

Billing

EC2

Group A-Z



Compute

EC2
EC2 Container Service
Lightsail [↗](#)
Elastic Beanstalk
Lambda
Batch



Storage

S3
EFS
Glacier
Storage Gateway



Database

RDS
DynamoDB
ElastiCache
Redshift



Networking & Content Delivery

VPC
CloudFront
Direct Connect
Route 53



Migration

Application Discovery Service
DMS
Server Migration
Snowball



Developer Tools

CodeCommit
CodeBuild
CodeDeploy
CodePipeline
X-Ray



Management Tools

CloudWatch
CloudFormation
CloudTrail
Config
OpsWorks
Service Catalog
Trusted Advisor
Managed Services



Security, Identity & Compliance

IAM
Inspector
Certificate Manager
Directory Service
WAF & Shield
Compliance Reports



Analytics

Athena
EMR
CloudSearch
Elasticsearch Service
Kinesis
Data Pipeline
QuickSight [↗](#)



Artificial Intelligence

Lex
Polly
Rekognition
Machine Learning



Internet Of Things

AWS IoT



Contact Center

Amazon Connect



Game Development

Amazon GameLift



Mobile Services

Mobile Hub
Cognito
Device Farm
Mobile Analytics
Pinpoint



Application Services

Step Functions
SWF
API Gateway
Elastic Transcoder



Messaging

Simple Queue Service
Simple Notification Service
SES



Business Productivity

WorkDocs
WorkMail
Amazon Chime [↗](#)



Desktop & App Streaming

WorkSpaces
AppStream 2.0

close



3 videos, 2 tutorials, 3 labs



3 videos, 5 tutorials, 3 labs



3 videos, 1 lab

Amazon EC2 Web Console

- ▶ AWS Console Home
- ▶ AWS Services
- ▶ AWS EC2 Console Home

- EC2 Dashboard
- Events
- Tags
- Reports
- Limits
- INSTANCES
- Instances
- Spot Requests
- Reserved Instances
- Scheduled Instances
- Dedicated Hosts
- IMAGES
- AMIs
- Bundle Tasks
- ELASTIC BLOCK STORE
- Volumes
- Snapshots
- NETWORK & SECURITY
- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network Interfaces
- LOAD BALANCING
- Load Balancers
- Target Groups
- AUTO SCALING
- Launch Configurations
- Auto Scaling Groups
- SYSTEMS MANAGER SERVICES
- Run Command
- Patch Compliance
- State Manager
- Automations
- Patch Baselines
- SYSTEMS MANAGER SHARED RESOURCES
- Managed Instances
- Activations
- Documents
- Maintenance Windows
- Parameter Store
- Patches

Resources

You are using the following Amazon EC2 resources in the US East (N. Virginia) region:

0 Running Instances	0 Elastic IPs
0 Dedicated Hosts	0 Snapshots
0 Volumes	0 Load Balancers
6 Key Pairs	31 Security Groups
3 Placement Groups	

Just need a simple virtual private server? Get everything you need to jumpstart your project - compute, storage, and networking - for a low, predictable price. [Try Amazon Lightsail for free.](#)

Create Instance

To start using Amazon EC2 you will want to launch a virtual server, known as an Amazon EC2 instance.

[Launch Instance](#)

Note: Your instances will launch in the US East (N. Virginia) region

Service Health

Service Status:

- US East (N. Virginia):
This service is operating normally

Availability Zone Status:

- us-east-1a:
Availability zone is operating normally
- us-east-1b:
Availability zone is operating normally
- us-east-1c:
Availability zone is operating normally
- us-east-1d:
Availability zone is operating normally
- us-east-1e:
Availability zone is operating normally

[Service Health Dashboard](#)

Scheduled Events

US East (N. Virginia):

No events

Account Attributes

Supported Platforms

VPC

Default VPC

vpc-57e3f333

Resource ID length management

Additional Information

[Getting Started Guide](#)

[Documentation](#)

[All EC2 Resources](#)

[Forums](#)

[Pricing](#)

[Contact Us](#)

AWS Marketplace

Find free software trial products in the AWS Marketplace from the [EC2 Launch Wizard](#). Or try these popular AMIs:

[Barracuda NextGen Firewall F-Series - PAYG](#)

Provided by Barracuda Networks, Inc.

Rating ★★★★★

Starting from \$0.60/hr or from \$4,599/yr (12% savings) for software + AWS usage fees

[View all Network Infrastructure](#)

[VM-Series Next-Generation Firewall Bundle 2](#)

Provided by Palo Alto Networks

Rating ★★★★★

\$1.28/hr or \$4,500/yr (60% savings) for software + AWS usage fees

[View all Security](#)

[ONTAP Cloud for AWS](#)

Provided by NetApp, Inc.

Rating ★★★★★

Starting from \$0.75/hr or from \$4,993/yr (24% savings) for software + AWS usage fees

[View all Storage](#)

[Find more software on AWS Marketplace](#)

Amazon EC2 Web Console

- ▶ AWS Console Home
- ▶ AWS Services
- ▶ AWS EC2 Console Home
- ▶ AWS EC2 Launch Instances

EC2 Dashboard

Events

Tags

Reports

Limits

- INSTANCES
- Instances
- Spot Requests
- Reserved Instances
- Scheduled Instances
- Dedicated Hosts
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- Patches

Launch Instance

Connect

Actions ▾

🔍 Filter by tags and attributes or search by keyword

🔍 < < None found > >

You do not have any running instances in this region.

First time using EC2?? Check out the [Getting Started Guide](#).

Click the Launch Instance button to start your own server.

Launch Instance

Select an instance above



Amazon EC2 Web Console

- ▶ AWS Console Home
- ▶ AWS Services
- ▶ AWS EC2 Console Home
- ▶ AWS EC2 Launch Instances
- ▶ Choose an AMI

Step 1: Choose an Amazon Machine Image (AMI)

[Cancel and Exit](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.

Quick Start

1 to 31 of 31 AMIs

My AMIs

AWS Marketplace

Community AMIs

☐ Free tier only ⓘ



Amazon Linux AMI 2017.03.0 (HVM), SSD Volume Type - ami-22ce4934

Amazon Linux

Free tier eligible

The Amazon Linux AMI is an EBS-backed, AWS-supported image. The default image includes AWS command line tools, Python, Ruby, Perl, and Java. The repositories include Docker, PHP, MySQL, PostgreSQL, and other packages.

Root device type: ebs Virtualization type: hvm

Select

64-bit



Red Hat

Free tier eligible

Red Hat Enterprise Linux 7.3 (HVM), SSD Volume Type - ami-b63769a1

Red Hat Enterprise Linux version 7.3 (HVM), EBS General Purpose (SSD) Volume Type

Root device type: ebs Virtualization type: hvm

Select

64-bit



SUSE Linux

Free tier eligible

SUSE Linux Enterprise Server 12 SP2 (HVM), SSD Volume Type - ami-fde4ebee

SUSE Linux Enterprise Server 12 Service Pack 2 (HVM), EBS General Purpose (SSD) Volume Type. Public Cloud, Advanced Systems Management, Web and Scripting, and Legacy modules enabled.

Root device type: ebs Virtualization type: hvm

Select

64-bit



Ubuntu

Free tier eligible

Ubuntu Server 16.04 LTS (HVM), SSD Volume Type - ami-f4cc1de2

Ubuntu Server 16.04 LTS (HVM), EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Root device type: ebs Virtualization type: hvm

Select

64-bit



Windows

Free tier eligible

Microsoft Windows Server 2016 Base - ami-b6af04a0

Microsoft Windows 2016 Datacenter edition, [English]

Root device type: ebs Virtualization type: hvm

Select

64-bit



Amazon RDS

Are you launching a database instance? Try Amazon RDS.

Amazon Relational Database Service (RDS) makes it easy to set up, operate, and scale your database on AWS by automating time-consuming database management tasks. With RDS, you can easily deploy **Amazon Aurora**, **MariaDB**, **MySQL**, **Oracle**, **PostgreSQL**, and **SQL Server** databases on AWS. **Aurora** is a MySQL-compatible, enterprise-class database at 1/10th the cost of commercial databases. [Learn more about RDS](#)

[Launch a database using RDS](#)

[Hide](#)



Windows

Free tier eligible

Microsoft Windows Server 2016 Base with Containers - ami-2dae053b

Microsoft Windows 2016 Datacenter edition with Containers, [English]

Root device type: ebs Virtualization type: hvm

Select

64-bit



Windows

Free tier eligible

Microsoft Windows Server 2016 Base Nano - ami-2931863f

Microsoft Windows 2016 Datacenter Edition Nano, [English]

Root device type: ebs Virtualization type: hvm

Select

64-bit



Windows

Microsoft Windows Server 2016 with SQL Server Express - ami-f7a209e1

Microsoft Windows 2016 Datacenter edition, Microsoft SQL Server 2016 Express, [English]

Root device type: ebs Virtualization type: hvm

Select

64-bit

Amazon EC2 Web Console

- ▶ AWS Console Home
- ▶ AWS Services
- ▶ AWS EC2 Console Home
- ▶ AWS EC2 Launch Instances
- ▶ Choose an AMI
- ▶ Choose an Instance Type

Step 2: Choose an Instance Type

Amazon EC2 provides a wide selection of instance types optimized to fit different use cases. Instances are virtual servers that can run applications. They have varying combinations of CPU, memory, storage, and networking capacity, and give you the flexibility to choose the appropriate mix of resources for your applications. [Learn more](#) about instance types and how they can meet your computing needs.

Filter by: All instance types Current generation Show/Hide Columns

Currently selected: t2.micro (Variable ECUs, 1 vCPUs, 2.5 GHz, Intel Xeon Family, 1 GiB memory, EBS only)

	Family	Type	vCPUs	Memory (GiB)	Instance Storage (GiB)	EBS-Optimized Available	Network Performance	IPv6 Support
☐	General purpose	t2.nano	1	0.5	EBS only	-	Low to Moderate	Yes
☒	General purpose	t2.micro Free tier eligible	1	1	EBS only	-	Low to Moderate	Yes
☐	General purpose	t2.small	1	2	EBS only	-	Low to Moderate	Yes
☐	General purpose	t2.medium	2	4	EBS only	-	Low to Moderate	Yes
☐	General purpose	t2.large	2	8	EBS only	-	Low to Moderate	Yes
☐	General purpose	t2.xlarge	4	16	EBS only	-	Moderate	Yes
☐	General purpose	t2.2xlarge	8	32	EBS only	-	Moderate	Yes
☐	General purpose	m4.large	2	8	EBS only	Yes	Moderate	Yes
☐	General purpose	m4.xlarge	4	16	EBS only	Yes	High	Yes
☐	General purpose	m4.2xlarge	8	32	EBS only	Yes	High	Yes
☐	General purpose	m4.4xlarge	16	64	EBS only	Yes	High	Yes
☐	General purpose	m4.10xlarge	40	160	EBS only	Yes	10 Gigabit	Yes
☐	General purpose	m4.16xlarge	64	256	EBS only	Yes	20 Gigabit	Yes
☐	General purpose	m3.medium	1	3.75	1 x 4 (SSD)	-	Moderate	-
☐	General purpose	m3.large	2	7.5	1 x 32 (SSD)	-	Moderate	-
☐	General purpose	m3.xlarge	4	15	2 x 40 (SSD)	Yes	High	-
☐	General purpose	m3.2xlarge	8	30	2 x 80 (SSD)	Yes	High	-
☐	Compute optimized	c4.large	2	3.75	EBS only	Yes	Moderate	Yes
☐	Compute optimized	c4.xlarge	4	7.5	EBS only	Yes	High	Yes
☐	Compute optimized	c4.2xlarge	8	15	EBS only	Yes	High	Yes
☐	Compute optimized	c4.4xlarge	16	30	EBS only	Yes	High	Yes

Amazon EC2 Web Console

- ▶ AWS Console Home
- ▶ AWS Services
- ▶ AWS EC2 Console Home
- ▶ AWS EC2 Launch Instances
- ▶ Choose an AMI
- ▶ Choose an Instance Type
- ▶ Review Instance Launch

Step 7: Review Instance Launch

Please review your instance launch details. You can go back to edit changes for each section. Click **Launch** to assign a key pair to your instance and complete the launch process.



Improve your instances' security. Your security group, launch-wizard-18, is open to the world.

Your instances may be accessible from any IP address. We recommend that you update your security group rules to allow access from known IP addresses only.

You can also open additional ports in your security group to facilitate access to the application or service you're running, e.g., HTTP (80) for web servers. [Edit security groups](#)

AMI Details

[Edit AMI](#)

Ubuntu Server 16.04 LTS (HVM), SSD Volume Type - ami-f4cc1de2

Free tier
eligible

Ubuntu Server 16.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Root Device Type: ebs Virtualization type: hvm

Instance Type

[Edit instance type](#)

Instance Type	ECUs	vCPUs	Memory (GiB)	Instance Storage (GB)	EBS-Optimized Available	Network Performance
t2.micro	Variable	1	1	EBS only	-	Low to Moderate

Security Groups

[Edit security groups](#)

Security group name
Description

launch-wizard-18
launch-wizard-18 created 2017-04-06T09:56:24.292+02:00

Type ⓘ	Protocol ⓘ	Port Range ⓘ	Source ⓘ
SSH	TCP	22	0.0.0.0/0

Instance Details

[Edit instance details](#)

Storage

[Edit storage](#)

Tags

[Edit tags](#)[Cancel](#)[Previous](#)[Launch](#)

Amazon EC2 Web Console

- ▶ AWS Console Home
- ▶ AWS Services
- ▶ AWS EC2 Console Home
- ▶ AWS EC2 Launch Instances
- ▶ Choose an AMI
- ▶ Choose an Instance Type
- ▶ Review Instance Launch
- ▶ Select an existing key pair or create a new key pair

Step 7: Review Instance Launch

Please review your instance launch details. You can go back to edit changes for each section. Click Launch to assign a key pair to your instance and complete the launch process.

 **Improve your instances' security. Your security group, launch-wizard-18, is open to the world.**
Your instances may be accessible from any IP address. We recommend that you update your security group rules to allow access from known IP addresses only.
You can also open additional ports in your security group to facilitate access to the application or service you're running, e.g., HTTP (80) for web servers. [Edit security groups](#)

AMI Details

[Edit AMI](#)

 **Ubuntu Server 16.04 LTS (HVM), SSD Volume Type - ami-f4cc1de2**

Free tier eligible

Ubuntu Server 16.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).
Root Device Type: ebs Virtualization type: hvm

Instance Type

[Edit instance type](#)

Instance Type	ECUs	vCPUs
t2.micro	Variable	1

Security Groups

[Edit security groups](#)

Security group name
Description

launch-wizard-18
launch-wizard-18 created 2

Type ⓘ

SSH

Prote

TCP

Instance Details

[Edit instance details](#)

Storage

[Edit storage](#)

Tags

[Edit tags](#)

Select an existing key pair or create a new key pair

✕

A key pair consists of a **public key** that AWS stores, and a **private key file** that you store. Together, they allow you to connect to your instance securely. For Windows AMIs, the private key file is required to obtain the password used to log into your instance. For Linux AMIs, the private key file allows you to securely SSH into your instance.

Note: The selected key pair will be added to the set of keys authorized for this instance. Learn more about [removing existing key pairs from a public AMI](#).

Create a new key pair

Key pair name

test

Download Key Pair

 You have to download the **private key file** (*.pem file) before you can continue. **Store it in a secure and accessible location.** You will not be able to download the file again after it's created.

Cancel

Launch Instances

Amazon EC2 Web Console

- ▶ Launch Status



Launch Status



Your instances are now launching

The following instance launches have been initiated: [i-0e47d6090737f083e](#) [View launch log](#)



Get notified of estimated charges

[Create billing alerts](#) to get an email notification when estimated charges on your AWS bill exceed an amount you define (for example, if you exceed the free usage tier).

How to connect to your instances

Your instances are launching, and it may take a few minutes until they are in the **running** state, when they will be ready for you to use. Usage hours on your new instances will start immediately and continue to accrue until you stop or terminate your instances.

Click **View Instances** to monitor your instances' status. Once your instances are in the **running** state, you can **connect** to them from the Instances screen. [Find out](#) how to connect to your instances.

▼ Here are some helpful resources to get you started

- [How to connect to your Linux instance](#)
- [Amazon EC2: User Guide](#)
- [Learn about AWS Free Usage Tier](#)
- [Amazon EC2: Discussion Forum](#)

While your instances are launching you can also

[Create status check alarms](#) to be notified when these instances fail status checks. (Additional charges may apply)

[Create and attach additional EBS volumes](#) (Additional charges may apply)

[Manage security groups](#)

[View Instances](#)

Amazon EC2 Web Console

- ▶ Launch Status
- ▶ Running Instances

- EC2 Dashboard
- Events
- Tags
- Reports
- Limits

- INSTANCES
 - Instances
 - Spot Requests
 - Reserved Instances
 - Scheduled Instances
 - Dedicated Hosts

- IMAGES
 - AMIs
 - Bundle Tasks

- ELASTIC BLOCK STORE
 - Volumes
 - Snapshots

- NETWORK & SECURITY
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 - Elastic IPs
 - Placement Groups
 - Key Pairs
 - Network Interfaces

- LOAD BALANCING
 - Load Balancers
 - Target Groups

- AUTO SCALING
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- SYSTEMS MANAGER SERVICES
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- SYSTEMS MANAGER SHARED RESOURCES
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Launch Instance Connect Actions

Filter by tags and attributes or search by keyword									
Name	Instance ID	Instance Type	Availability Zone	Instance State	Status Checks	Alarm Status	Public DNS (IPv4)	IPv4 Public IP	IPv6 IPs
	i-0e47d6090737f083e	t2.micro	us-east-1c	running	Initializing	None	ec2-54-236-57-16.comp...	54.236.57.16	-

Instance: i-0e47d6090737f083e Public DNS: ec2-54-236-57-16.compute-1.amazonaws.com			
Description Status Checks Monitoring Tags			
Instance ID	i-0e47d6090737f083e	Public DNS (IPv4)	ec2-54-236-57-16.compute-1.amazonaws.com
Instance state	running	IPv4 Public IP	54.236.57.16
Instance type	t2.micro	IPv6 IPs	-
Elastic IPs		Private DNS	ip-172-31-65-243.ec2.internal
Availability zone	us-east-1c	Private IPs	172.31.65.243
Security groups	launch-wizard-18, view inbound rules	Secondary private IPs	
Scheduled events	No scheduled events	VPC ID	vpc-67e3ff33
AMI ID	ubuntu/images/hvm-ssd/ubuntu-xenial-16.04-amd64-server-20170221 (ami-f4cc1de2)	Subnet ID	subnet-d03d5eb8
Platform	-	Network interfaces	eth0
IAM role	-	Source/dest. check	True
Key pair name	test4		

Amazon EC2 Web Console

- ▶ Launch Status
- ▶ Running Instances
- ▶ Options of an Instance

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Activations
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Launch Instance Connect Actions ▾

Filter by tags and attributes or search by keyword

Name	Instance ID	Instance Type	Availability Zone	Instance State	Status Checks	Alarm Status	Public DNS (IPv4)	IPv4 Public IP	IPv6 IPs
	i-0e47d6090737f083e	t2.micro		running	Initializing	None	ec2-54-236-57-16.comp...	54.236.57.16	-

- Connect
 - Get Windows Password
 - Launch More Like This
- Instance State ▸
- Instance Settings ▸
- Image ▸
- Networking ▸
- CloudWatch Monitoring ▸

Instance: i-0e47d6090737f083e Public DNS: ec2-54-236-57-16.compute-1.amazonaws.com

Description Status Checks Monitoring Tags

Instance ID	i-0e47d6090737f083e	Public DNS (IPv4)	ec2-54-236-57-16.compute-1.amazonaws.com
Instance state	running	IPv4 Public IP	54.236.57.16
Instance type	t2.micro	IPv6 IPs	-
Elastic IPs		Private DNS	ip-172-31-65-243.ec2.internal
Availability zone	us-east-1c	Private IPs	172.31.65.243
Security groups	launch-wizard-18, view inbound rules	Secondary private IPs	
Scheduled events	No scheduled events	VPC ID	vpc-57e3#33
AMI ID	ubuntu/images/hvm-ssd/ubuntu-xenial-16.04-amd64-server-20170221 (ami-14cc1de2)	Subnet ID	subnet-dd3d5eb8
Platform	-	Network interfaces	eth0
IAM role	-	Source/dest. check	True
Key pair name	test4		

Amazon EC2 Web Console

- ▶ Launch Status
- ▶ Running Instances
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- ▶ Connect to an Instance

Connect To Your Instance



I would like to connect with

- ☒ A standalone SSH client
☐ A Java SSH Client directly from my browser (Java required)

To access your instance:

1. Open an SSH client. (find out how to [connect using PuTTY](#))
2. Locate your private key file (test4.pem). The wizard automatically detects the key you used to launch the instance.
3. Your key must not be publicly viewable for SSH to work. Use this command if needed:

```
chmod 400 test4.pem
```

4. Connect to your instance using its Public DNS:

```
ec2-54-236-57-16.compute-1.amazonaws.com
```

Example:

```
ssh -i "test4.pem" ubuntu@ec2-54-236-57-16.compute-1.amazonaws.com
```

Please note that in most cases the username above will be correct, however please ensure that you read your AMI usage instructions to ensure that the AMI owner has not changed the default AMI username.

If you need any assistance connecting to your instance, please see our [connection documentation](#).

Close

StarCluster HPC AMI Amazon EC2

StarCluster

HPC AMI

- ▶ Where is the StarCluster AMI for HPC:
 - ▶ AMI ID ami-52a0c53b (HVM+GPU) based on Ubuntu 12.04;
 - ▶ available on N.Virginia region (us-east-1);
 - ▶ for all available StarCluster AMIs execute the command
`starcluster --region sa-east-1 listpublic`.

StarCluster

HPC AMI softwares

- ▶ The AMIs include the following software:
 - ▶ OpenMPI - Library used for writing/running parallel applications.
 - ▶ Hadoop - Framework for writing massively distributed map/reduce tasks.
 - ▶ Open Grid Scheduler - Queueing system for scheduling jobs on the cluster and handling load balancing.
 - ▶ NFS - Network File System for sharing folders across the cluster.
 - ▶ ATLAS - Automatically Tuned Linear Algebra Subroutines (custom built for larger EC2 instance types).
 - ▶ Scipy - Scientific algorithms library for Python (compiled against custom Atlas).
 - ▶ Numpy - Fast array and numerical library for Python (compiled against custom Atlas).
 - ▶ IPython - An advanced interactive shell for Python.

Thank you for your attention . . .

Next Lesson Amazon Web Services: Working with an MPI Cluster



ISISLab